



Advanced Level **Biology Notes**

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FORM 5 - TERM 1

TOPIC 2: BIOLOGICAL MOLECULES AND WATER.

KEY CONCEPT.

..... **CARBOHYDRATES.**

OBJECTIVES.

- *Carry out tests to identify carbohydrates*
- *Describe the formation and breakage of glycosidic bond.*
- *Describe the synthesis and molecular structure of polysaccharides*
- *Relate structures of polysaccharides to their functions in living organisms*

The whole of nature, as has been said, is a conjugation of the verb to eat, in the active and passive.
—William Ralph Inge

A macromolecule is a large biological molecule such as a protein, polysaccharide or nucleic acid.

A monomer is a relatively simple molecule which is used as a basic building block for the synthesis of a polymer; many monomers are joined together to make the polymer, usually by condensation reactions; common examples of molecules used as monomers are monosaccharides, amino acids and nucleotides.

A polymer is a giant molecule made from many similar repeating subunits joined together in a chain; the subunits are much smaller and simpler molecules known as monomers; examples of biological polymers are polysaccharides, proteins and nucleic acids.

A monosaccharide is a molecule consisting of a single sugar unit with the general formula $(CH_2O)_n$.

A disaccharide is a sugar molecule consisting of two monosaccharides joined together by a glycosidic bond.

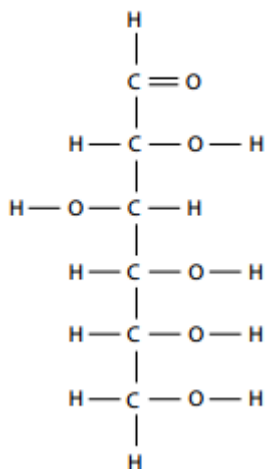
A polysaccharide is a polymer whose subunits are monosaccharides joined together by glycosidic bonds.

CARBOHYDRATES

- All carbohydrates contain the elements carbon, hydrogen and oxygen.
- The 'hydrate' part of the name comes from the fact that hydrogen and oxygen atoms are present in the ratio of 2:1, as they are in water ('hydrate' refers to water).
- The general formula for a carbohydrate can therefore be written as $C_x(H_2O)_y$.
- Carbohydrates are divided into three main groups, namely monosaccharides, disaccharides and polysaccharides.
- The word 'saccharide' refers to a sugar or sweet substance.

MONOSACCHARIDES

- Monosaccharides are sugars.
- Sugars dissolve easily in water to form sweet-tasting solutions.
- Monosaccharides have the general formula $(CH_2O)_n$ and consist of a single sugar molecule ('mono' means one).
- The main types of monosaccharides, if classified according to the number of carbon atoms in each molecule, are trioses(3C), pentoses (5C) and hexoses (6C).
- The names of all sugars end with -ose.
- Common hexoses are glucose, fructose and galactose.
- Two common pentoses are ribose and deoxyribose.
- The formula for a hexose can be written as $C_6H_{12}O_6$ and this is known as the molecular formula.
- A diagram known as the structural formula shows the arrangement of atoms in a molecule.



- A significant feature of the structures of pentoses and hexoses is that they intra-bond and form more stable ring structures.
- When glucose forms a ring, carbon atom number 1 joins to the oxygen on carbon atom number 5.
- The ring therefore contains oxygen, and carbon atom number 6 is not part of the ring.
- The hydroxyl group, $-OH$, on carbon atom 1 may be above or below the plane of the ring and thus glucose has two isomers.
- The isomer of glucose where the $-OH$ is below the ring is known as α -glucose and the isomer where it is above the ring is β -glucose.
- The existence of α - and β -isomers has important biological consequences in Biochemistry.

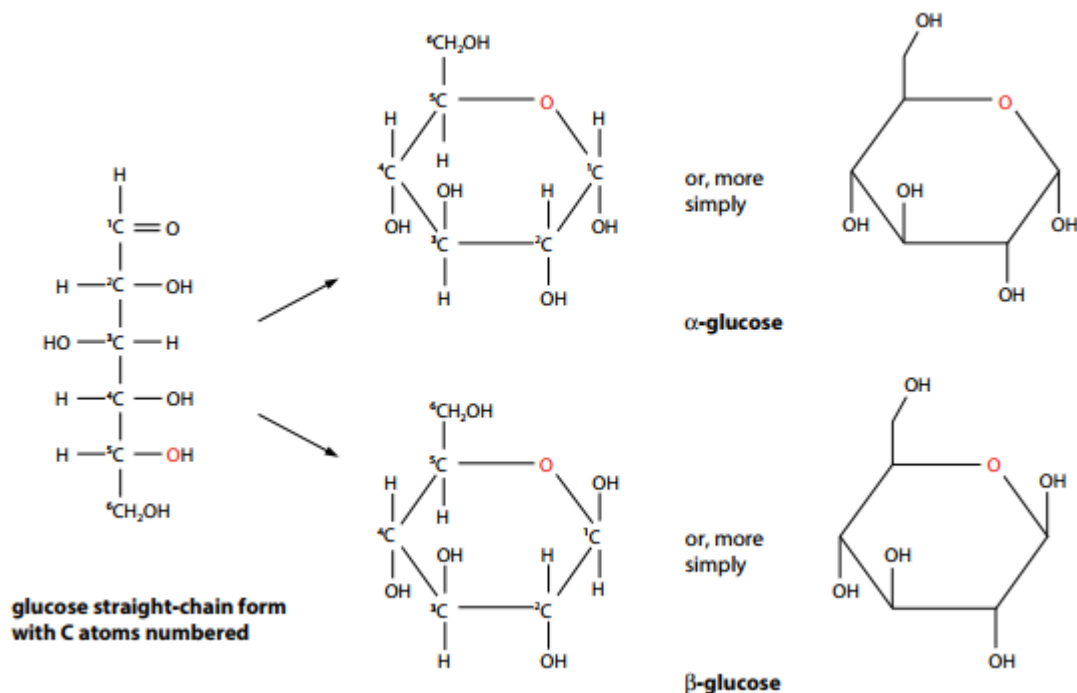
FUNCTIONS OF MONOSACCHARIDES

1. Source of energy in respiration.

- ✚ The large number of carbon–hydrogen bonds can be broken to release a lot of energy, which is used to make ATP from ADP and phosphate.
- ✚ The most important monosaccharide in energy metabolism is glucose.

2. Subunits of organic molecules.

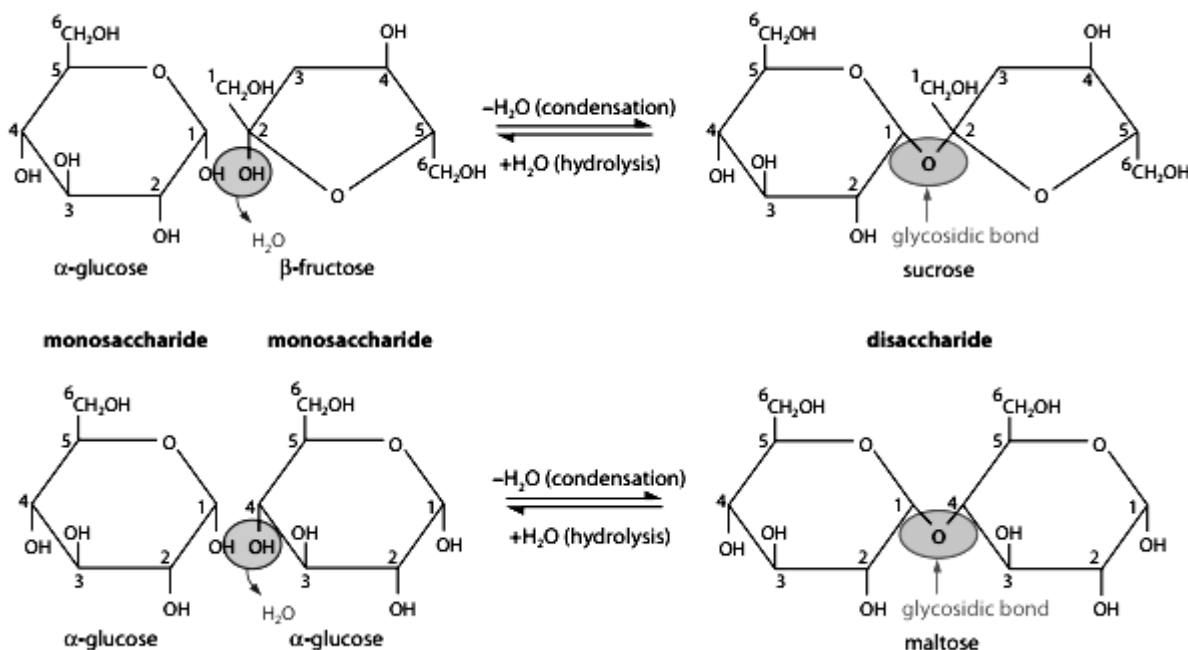
- ✚ Secondly, monosaccharides are important as building blocks for large molecules.
- ✚ For example, glucose is used to make the polysaccharides starch, glycogen and cellulose.
- ✚ Ribose (a pentose) is one of the molecules used to make RNA (ribonucleic acid) and ATP. Deoxyribose (also a pentose) is one of the molecules used to make DNA.



DISACCHARIDES.

- ✚ Disaccharides are sugars.
- ✚ They are formed by two monosaccharides joining together in a condensation reaction.
- ✚ In each condensation reaction, two hydroxyl ($-OH$) groups are oriented alongside each other.
- ✚ One $-OH$ combines with a hydrogen atom from the other to form a water molecule.
- ✚ A bridge called a glycosidic bond forms between the two molecules, holding them together.
- ✚ The enzyme controlling the condensation reaction determines which $-OH$ groups come alongside each other.
- ✚ The reverse of condensation is the addition of water, which is known as hydrolysis.
- ✚ This takes place during the digestion of disaccharides and polysaccharides, when they are broken down to monosaccharides.
- ✚ The three most common disaccharides are:
 1. Maltose (glucose+ glucose).
 2. Sucrose (glucose+ fructose).
 3. Lactose (glucose + galactose).

***NB: Notice that fructose has a different ring structure to glucose.**

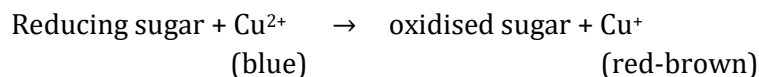


USES OF DI-SACCHARIDES

1. They are broken down to monosaccharides to supply energy to body cells.
2. Lactose in milk provides infant animals with their main source of energy as it is broken down to the monosaccharides glucose and galactose.
3. Sucrose is soluble and is the transport sugar in the phloem sap most vascular plants.

REDUCING SUGARS AND NON-REDUCING SUGARS

- ✚ Reducing sugars include all monosaccharides, such as glucose, and some disaccharides, such as maltose.
- ✚ The only common non-reducing sugar is sucrose.
- ✚ Reducing sugars reduce certain atoms and are oxidised in the process.
- ✚ Benedict's reagent is copper (II) sulphate in an alkaline solution and has a blue colour.
- ✚ Reducing sugars reduce soluble blue copper sulphate, containing copper (II) ions, to insoluble brick-red copper oxide, containing copper (I).
- ✚ The copper oxide is observed as a brick-red precipitate.
- ✚ Sucrose is not a reducing sugar, so you would get a negative result from Benedict's test.
- ✚ Such a disaccharide like sucrose is first broken down into its two monosaccharide constituents.
- ✚ The chemical reaction is hydrolysis and can be done using hydrochloric acid.
- ✚ The constituent monosaccharides are reducing sugars and their presence can be tested for using Benedict's test after the acid has been neutralised.



POLYSACCHARIDES

- ✚ Polysaccharides are polymers whose monomers are monosaccharides.
- ✚ They are made by joining many monosaccharide molecules through condensation.
- ✚ Each successive monosaccharide is held to the chain by means of a glycosidic bond.
- ✚ The final molecule may be several thousand monosaccharide units long, forming a macromolecule.
- ✚ The most important storage polysaccharides are starch and glycogen.
- ✚ Cellulose is the most important structural polysaccharide and is the most abundant.
- ✚ Polysaccharides are not sugars.

WHY SYNTHESIZE POLYSACCHARIDES?

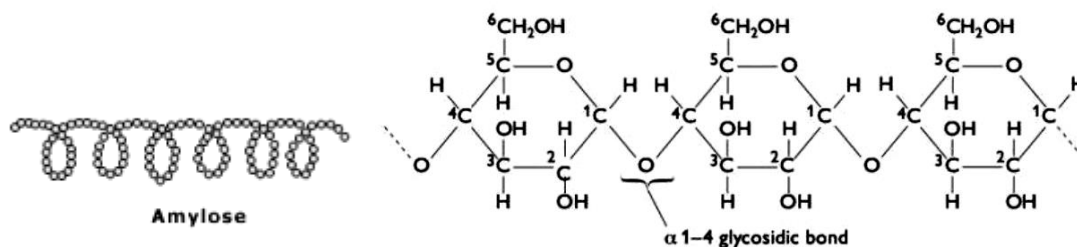
- ✚ Glucose is the main source of energy and has to be stored in an appropriate form.
- ✚ If glucose itself accumulated in cells, it would dissolve and make the contents of the cell too concentrated, which would affect its osmotic properties.
- ✚ Glucose is a reactive molecule and would interfere with normal cell chemistry.
- ✚ Glucose can be stored, by condensation reactions, to a storage polysaccharide, which is a convenient, compact, unreactive and insoluble molecule.
- ✚ The large sizes of these storage molecules are also not easily transported across the membrane therefore they can be retained within cells.
- ✚ The storage polysaccharide formed is starch in plants and glycogen in animals.
- ✚ Glucose can be made available again quickly by enzyme-controlled hydrolysis.

STARCH

- Natural starches contain nearly 20% amylose and 80% amylopectin.
- Large starch grains are commonly found in chloroplasts and in storage organs such as potato tubers.
- Starch grains are easily seen with a light microscope, especially if stained with iodine solution.
- Starch amylose molecules tend to curl up into long spirals.
- To test for starch, iodine solution is used.
- Iodine does not dissolve in water, so the 'iodine solution' is potassium iodide solution.
- The hole that runs down the middle of amylose spirals allows iodine molecules to fit into it.
- The starch-iodine complex that forms has a strong blue-black colour.

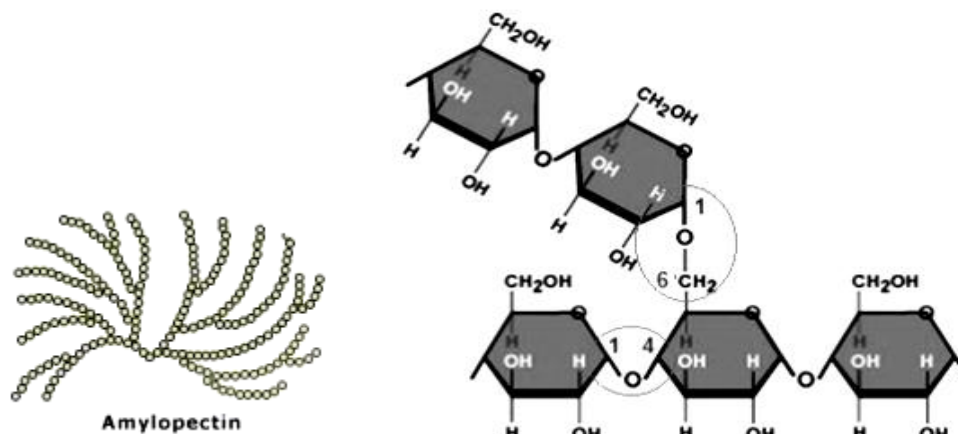
AMYLOSE

- Amylose is made by condensations between α -glucose molecules.
- A long, unbranching chain of 1,4 linked glucose molecules is constructed.
- The chains are coiled up into helical structures like springs, making the final molecule compact.



AMYLOPECTIN

- Amylopectin is also made of many 1,4 linked α -glucose molecules, but the chains are shorter than in amylose, and branch out to the sides.
- The branches are formed by 1,6 linkages.

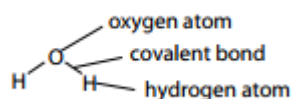


GLYCOGEN

- ✚ Starch is never found in animal cells.
- ✚ Instead, a substance with molecules very like those of amylopectin is used as the storage carbohydrate.
- ✚ This is called glycogen.
- ✚ Glycogen, like amylopectin, is made of chains of 1,4 linked α -glucose with 1,6 linkages forming branches.
- ✚ Glycogen molecules tend to be even-more branched than amylopectin molecules.
- ✚ Glycogen molecules clump together to form granules, which are visible in liver cells and muscle cells, where they form an energy reserve.

HYDROGEN BONDING.

- When atoms in molecules are held together by covalent bonds, they share electrons with each other forming one covalent bond.

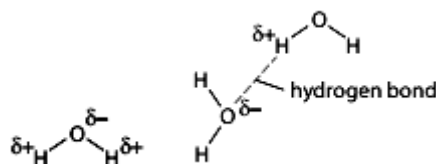


- However, sometimes the electrons are not shared equally.
- In water, the oxygen atom slightly draws electrons to itself more, and so has a small negative charge, written δ^- (delta minus).
- The electrons become further away from hydrogen atoms creating a small positive charge, written δ^+ (delta plus).
- This unequal distribution of charge is called a dipole or polarity.
- In water, the negatively charged oxygen of one molecule is attracted to a positively charged hydrogen of another, and this attraction is called a hydrogen bond.

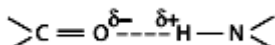
*NB

Notice that all Hydrogen bonds are always drawn using a dotted line.

- It is much weaker than a covalent bond, but still has a very significant effect.

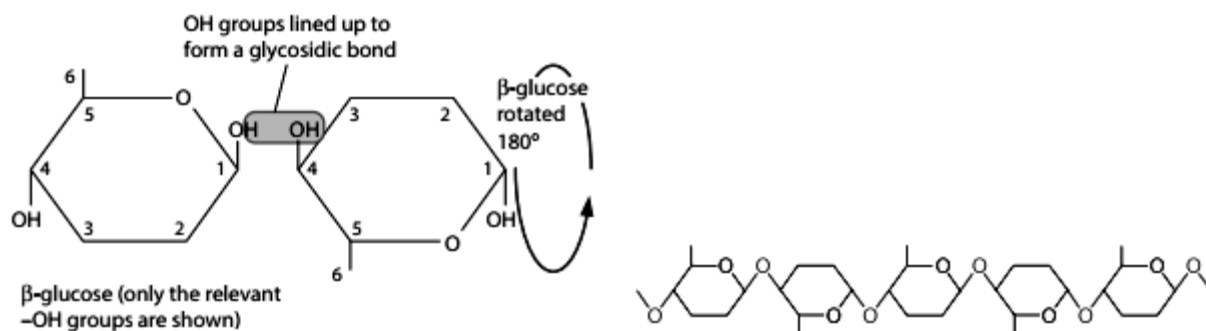


- Dipoles occur in many different molecules, particularly wherever there is an $-OH$, $-CO$ or $-NH$ group hence Hydrogen bonds can form between these groups.
- Molecules which have groups with dipoles, such as sugars, are said to be polar.
- They are attracted to water molecules, because the water molecules also have dipoles.
- Such molecules are said to be hydrophilic (water-loving), and they tend to be soluble in water.
- Molecules which do not have dipoles are said to be non-polar.
- They are not attracted to water, and they are hydrophobic (water-hating).



CELLULOSE

- Cellulose has a structural role, because it is a mechanically strong molecule.
- Cellulose is a polymer of β -glucose, not α -glucose.
- In the β -isomer, the -OH group on carbon atom 1 projects above the ring.
- In order to form a glycosidic bond with carbon atom 4, one glucose molecule must be rotated 180° relative to the other.
- Successive glucose units are linked at 180° to each other.
- This arrangement of β -glucose molecules results in strong molecules because the hydrogen atoms of -OH groups are weakly attracted to oxygen atoms in the same cellulose molecule resulting in attraction forces known as cohesion due to Hydrogen bonds.
- Hydrogen bonds are individually weak, but so many can form, due to the large number of -OH groups, that collectively they provide enormous strength.
- Cellulose molecules become tightly cross-linked to form bundles called micro-fibrils.
- Microfibrils are in turn held together in bundles called fibres by hydrogen bonding.



THE CELL WALL

- A cell wall typically has several layers of fibres, running in different directions to increase strength.
- Cellulose makes up about 30% of the average cell wall as other molecules help to cross-link the cellulose fibres, and some form a glue-like matrix around the fibres, which further increases strength.
- Cellulose fibres have a very high tensile strength hence they are very difficult to stretch or break making it possible for a cell to withstand the large pressures that develop within it as a result of osmosis.
- Without the wall, the cell would burst when in a dilute solution.
- Despite their strength, cellulose fibres are freely permeable, allowing water and solutes to reach or leave the cell surface membrane.

KEY CONCEPT**LIPIDS****OBJECTIVES**

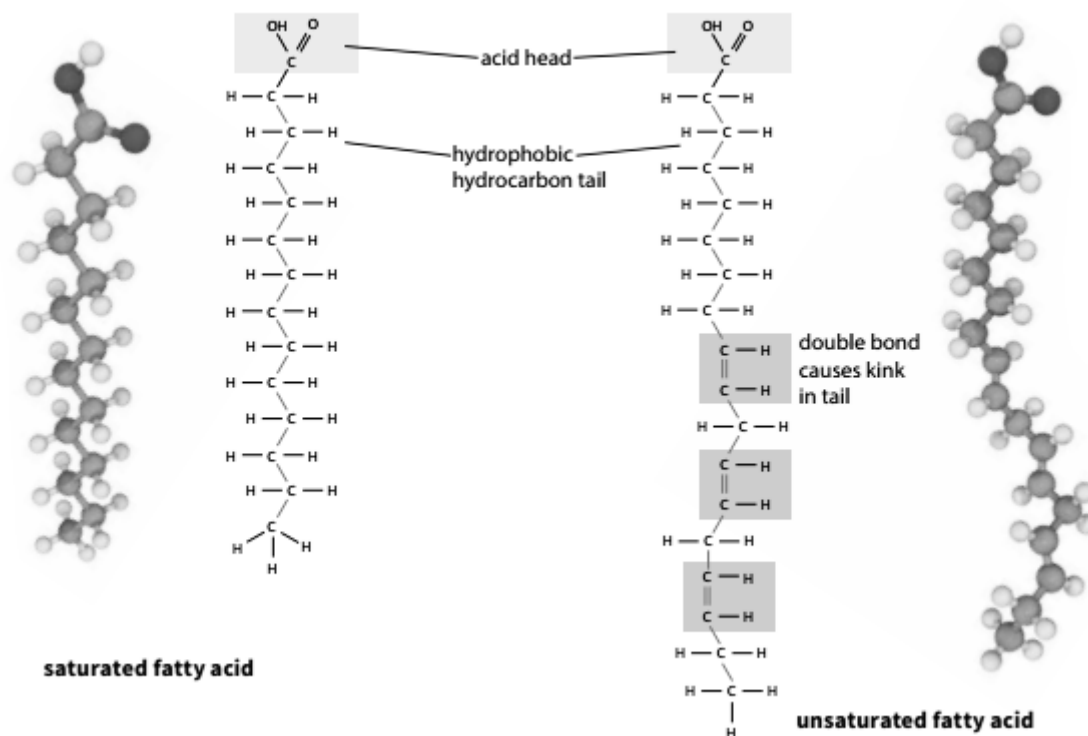
- *Identify lipids indifferent substances.*
- *Describe the molecular structures of a triglyceride and a phospholipid.*
- *Relate the structures of triglycerides and phospholipids to their functions in living organisms.*

LIPIDS

- ✚ True lipids are esters formed by fatty acids combining with an alcohol.
- ✚ Lipids are a very varied group of chemicals.
- ✚ They are all organic molecules which are insoluble in water.
- ✚ The most familiar lipids are fats and oils.
- ✚ Fats are solid at room temperature and oils are liquid at room temperature– chemically they are very similar.

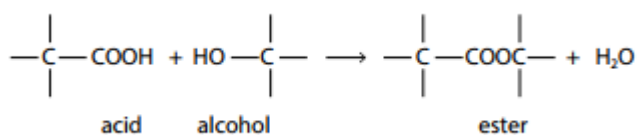
FATTY ACIDS

- ✚ Fatty acids contain the acidic group -COOH , known as a carboxyl group.
- ✚ These molecules have long hydrocarbon tails attached to the acidic 'head' of the molecule.
- ✚ The hydrocarbon tails consist of a chain of carbon atoms combined with hydrogen.
- ✚ The tails of some fatty acids have double bonds between neighbouring carbon atoms, like this:
 -C=C- forming unsaturated fatty acids as they do not contain the maximum possible amount of hydrogen.
- ✚ Double bonds lower the boiling point of lipids and most oils are unsaturated.
- ✚ If there is more than one double bond, the lipid is described as polyunsaturated; if there is only one it is monounsaturated.
- ✚ Animal lipids are often saturated and occur as fats, whereas plant lipids are often unsaturated and occur as oils, such as olive oil and sunflower oil.



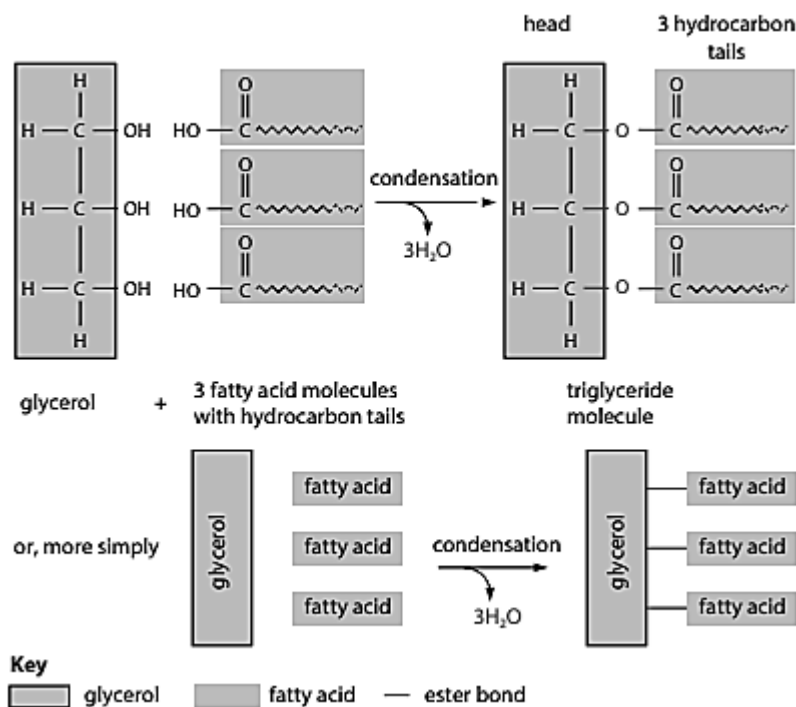
Alcohols

- Alcohols are a series of organic molecules which contain a hydroxyl group, $-\text{OH}$, attached to a carbon atom.
- Glycerol is an alcohol with three hydroxyl groups.
- The reaction between an acid and an alcohol produces a chemical known as an ester.
- The chemical link between the acid and the alcohol is known as an ester bond or an ester linkage.
- The $-\text{COOH}$ group on the acid reacts with the $-\text{OH}$ group on the alcohol to form the ester bond, $-\text{COO}-$.
- This is a condensation reaction because water is formed as a product.
- The resulting ester can be converted back to acid and alcohol by the reverse reaction of adding water, a reaction known as hydrolysis.



TRI-GLYCERIDES

- These are fats and oils.
- A glyceride is an ester formed by a fatty acid combining with the alcohol glycerol.
- Glycerol has three hydroxyl groups and each one is able to undergo a condensation reaction with a fatty acid.
- Triglycerides contain three fatty acids tails and three ester bonds.
- The tails can vary in length, depending on the fatty acids used.
- Triglycerides are insoluble in water but are soluble in certain organic solvents, including ether, chloroform and ethanol.
- This is because of the non-polar nature of the hydrocarbon tails: they have no uneven distribution of electrical charge.
- Hence, they are described as hydrophobic(water-hating).



FUNCTIONS OF LIPIDS

- Storage**
Triglycerides are insoluble in water and this makes them useful for storage in cells as they do not affect the osmotic properties of cells.
- Energy reserves**
The large number of Hydrogen atoms within lipids makes them high energy molecules hence they are excellent energy reserves because they release large amounts of energy when they are broken down. Animals e.g. bears that hibernate during cold winter months use water from the breakdown of stored fat for their body processes.
- Sources of water**
Lipids have many Hydrogen atoms hence when lipid molecules are oxidised during respiration, they release water. This occurs in camels and the desert Kangaroo.

4. Insulation

Fat stored under the skin in adipose tissue acts as an insulator against loss of body heat. Animals such as Penguins that live in very cold conditions have thick fat under their skins. In whales, the layer of fat is called blubber and gives the animals buoyancy as well as warmth.

5. Protection

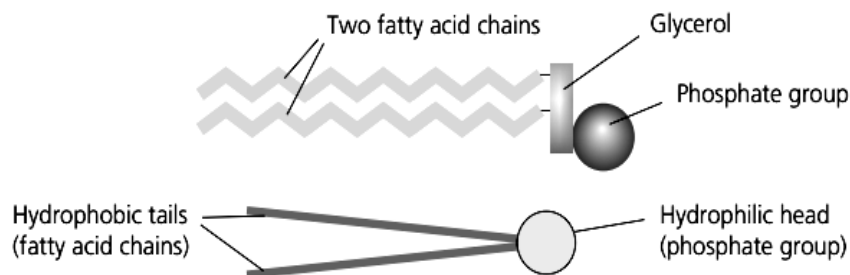
There is a thick layer of fat surrounding vital organs such as the heart and the kidneys to cushion and protect them from mechanical shock.

6. Absorption

Vitamin A and D are soluble in fats. Because of this, they can be absorbed through the wall of the small intestines into the blood.

PHOSPHOLIPIDS

- ✚ Phospholipids are a special type of lipid with the unusual property of having one end which is soluble in water and the rest of the molecule being insoluble in water.
- ✚ This is because one of the three fatty acid molecules is replaced by a phosphate group, which is polar.
- ✚ The phosphate group is hydrophilic (water-loving) and makes the head of a phospholipid molecule hydrophilic, although the two remaining tails are still hydrophobic.
- ✚ This allows the molecules to form a membrane around a cell, where the hydrophilic heads lie in the aqueous environment on the outside of the membrane, and the hydrophobic tails form a layer that is impermeable to polar substances.



KEY CONCEPT.

PROTEINS

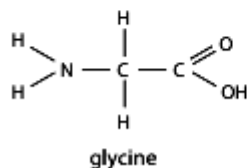
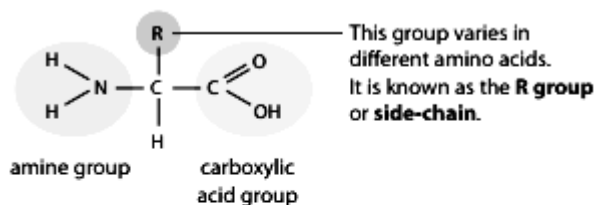
OBJECTIVES

- Describe the structure of an amino acid
- Outline the formation and breakage of a peptide bond
- Explain the meaning of primary, secondary, tertiary and quaternary structures of proteins
- Describe the types of bonds which hold the protein molecules in shape
- Describe the molecular structures of haemoglobin and collagen
- Relate the structures of haemoglobin and collagen to their functions in living organisms

- Proteins are an extremely important class of macromolecule in living organisms.
- Despite their tremendous range of functions, all proteins are made from the same basic monomers.
- These monomers are amino acids.

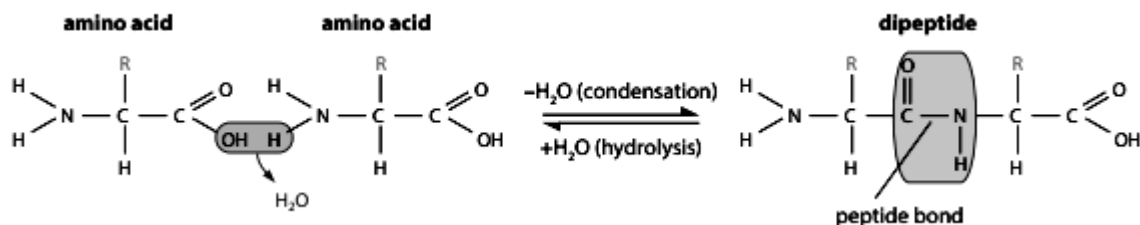
AMINO ACIDS

- They all have a central carbon atom which is bonded to an amine group, -NH_2 , and a carboxylic acid group, -COOH .
- A hydrogen atom is always bonded to the central carbon known as the alpha carbon atom.
- The only way in which amino acids differ from each other is in the other group of atoms bonded to the alpha carbon called the R group.
- There are 20 different amino acids which occur in the proteins of living organisms, all with a different R group.



DI-PEPTIDES

- During the formation of a di-peptide, one amino acid loses a hydroxyl ($-OH$) group from its carboxylic acid group, while the other loses a hydrogen atom from its amine group.
- This leaves a carbon atom of the first amino acid free to bond with the nitrogen atom of the second forming a peptide bond.
- The reaction is an example of a condensation reaction.



POLYPEPTIDES

- A molecule made up of many amino acids linked together by peptide bonds is called a polypeptide.
- A protein molecule may contain one polypeptide chain, or it may have more chains which interact with each other.
- In living cells, ribosomes are the sites where amino-acids are polymerized into polypeptides in enzyme-controlled polymerization reactions.
- Polypeptides can be broken down to amino acids by breaking the peptide bonds.
- This is a hydrolysis reaction, involving the addition of water, and happens naturally in the stomach and small intestine during digestion.

PRIMARY STRUCTURE OF A PROTEIN

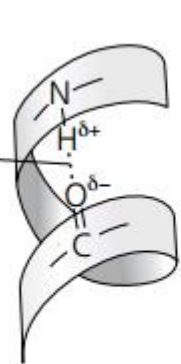
- The sequence in which specific amino acids are joined together in a polypeptide chain, is called the primary structure of the protein.
Val - Leu - Ser - Pro - Ala - Asp - Lys - Thr - Asn - Val - Lys - Ala
- There are an enormous number of different possible primary structures.
- A change in one amino acid in a protein molecule may alter the properties of the polypeptide.

SECONDARY STRUCTURE OF A PROTEIN

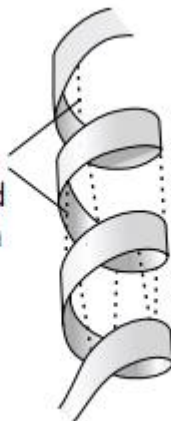
- It is the structure of a protein molecule resulting from the regular coiling or folding of the chain of amino acids, e.g. an α -helix or β -pleated sheet.
- A polypeptide chain, often coils into a corkscrew shape called an α -helix.
- Examples of proteins which have an α -helical shape include Keratin and Myosin.
- This secondary structure is due to hydrogen bonding between the oxygen of the $-CO-$ group of one amino acid and the hydrogen of the $-NH-$ group of the amino acid four places ahead of it.
- Sometimes hydrogen bonding can result in a much looser, straighter shape than the α -helix, which is called a β -pleated sheet.
- Examples of proteins that form the β -pleated sheet include the Fatty acid binding proteins (FABP).
- Hydrogen bonds, although strong enough to hold the α -helix and β -pleated sheet structures in shape, are easily broken by high temperatures and pH changes.
- In diagrams of protein structure, α -helices can be represented as coils or cylinders, β -sheets as arrows, and random coils as ribbon.

An alpha helix

Hydrogen bond between amino acids



Hydrogen bonds hold the helix in shape



A beta-pleated strand

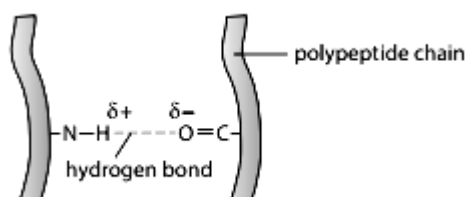


TERTIARY STRUCTURE OF A PROTEIN

- It is the compact structure of a protein molecule resulting from the three-dimensional coiling of the already-folded chain of amino acids.
- Myoglobin and Lysozyme are good examples of proteins with a tertiary structure.
- At first sight, the myoglobin and lysozyme molecules look like disorganised tangles, but this is not so as the shape of the molecules is very precise.
- The molecules are held in these exact shapes by bonds between amino acids in different parts of the chain.
- Four types of bonds help to keep folded proteins in their precise shapes and these are:

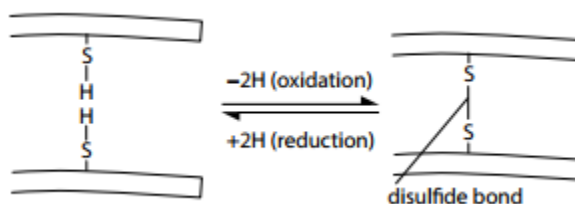
1. Hydrogen bonds

- Form between strongly polar groups –for example, –NH–, –CO– and –OH groups.
- Hydrogen bonds are easily broken by high temperatures and pH changes.



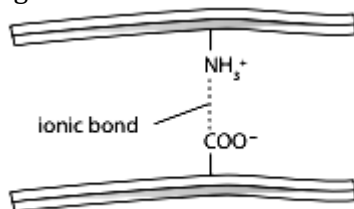
2. Disulphide bonds

- Form between two cysteine molecules and other amino acids which contain sulphur atoms.
- They are strong covalent bonds.
- They can be broken by reducing agents.



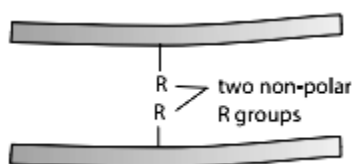
3. Ionic bonds

- form between R groups containing ionised amine (NH_3^+) groups and ionised carboxylic acid (COO^-) groups.
- They can be broken by pH changes



4. Hydrophobic interactions

- Occur between R groups which are non-polar, or hydrophobic.
- Although the interactions are weak, the groups tend to stay together because they are repelled by the watery environment around them.



QUATERNARY STRUCTURE OF A PROTEIN

- It is the three-dimensional arrangement of two or more polypeptides, or of a polypeptide and a non-protein component such as haem, in a protein molecule.
- Haemoglobin is an example of a protein with a Quaternary structure, having four polypeptide chains in each molecule.
- The chains are held together by the same four types of bonds as in the tertiary structure.

GLOBULAR PROTEINS

- Globular proteins have molecules that fold into a roughly spherical three-dimensional shape.
- Examples include Haemoglobin, Insulin and Enzymes.
- They are often soluble in water and may be metabolically active.

FIBROUS PROTEINS

- Fibrous proteins have long, thin molecules which often lie side by side to form fibres.
- Examples include Keratin and Collagen.
- They are not soluble and also are not metabolically active.
- They often have structural roles.

HAEMOGLOBIN

- ✚ Haemoglobin is a globular oxygen-carrying protein pigment found in red blood cells.
- ✚ It has a quaternary structure made up of four polypeptide chains.
- ✚ Each chain is known as globin.
- ✚ Two types of globins make up the human adult haemoglobin namely alpha-globin (α -globin) and beta-globin (β -globin).
- ✚ Two of the globin chains, called α chains, are made from α -globin, and the other two chains, called β chains, are made from β -globin.
- ✚ The haemoglobin molecule is nearly spherical.
- ✚ Each polypeptide chain of haemoglobin contains a haem group.
- ✚ A group like this, which is an important, permanent, part of a protein molecule but is not made of amino acids, is called a prosthetic group.
- ✚ Each haem group contains an iron atom.
- ✚ One oxygen molecule, O_2 can bind with each iron atom.
- ✚ A complete haemoglobin molecule, with four haem groups, can carry four oxygen molecules (eight oxygen atoms) at a time.
- ✚ It is the haem group which is responsible for the colour of haemoglobin.
- ✚ This colour changes depending on whether or not the iron atoms are combined with oxygen.
- ✚ If they are, the molecule is known as oxyhaemoglobin, and is bright red and if not, the colour is purplish.

RELATIONSHIP BETWEEN STRUCTURE AND FUNCTION IN HAEMOGLOBIN

1. Solubility

- ✚ The tertiary structure of haemoglobin makes it soluble.
- ✚ The four polypeptide chains are coiled up so that R groups with small charges on them (hydrophilic groups) are on the outside of the molecule.
- ✚ They therefore form hydrogen bonds with water molecules.
- ✚ Hydrophobic R groups are mostly found inside the molecule.

2. Ability to transport Carbon Dioxide

- ✚ 10% of CO_2 is transported bound to haemoglobin.
- ✚ Because CO_2 binds to the protein portion of haemoglobin, and not to the iron atoms of the haem groups, it does not compete with oxygen.
- ✚ However, it does cause the haemoglobin shape to change, lowering its affinity for oxygen.
- ✚ This contributes to the release of Oxygen in actively respiring tissues in what is known as the Bohr effect.

3. Buffering ability

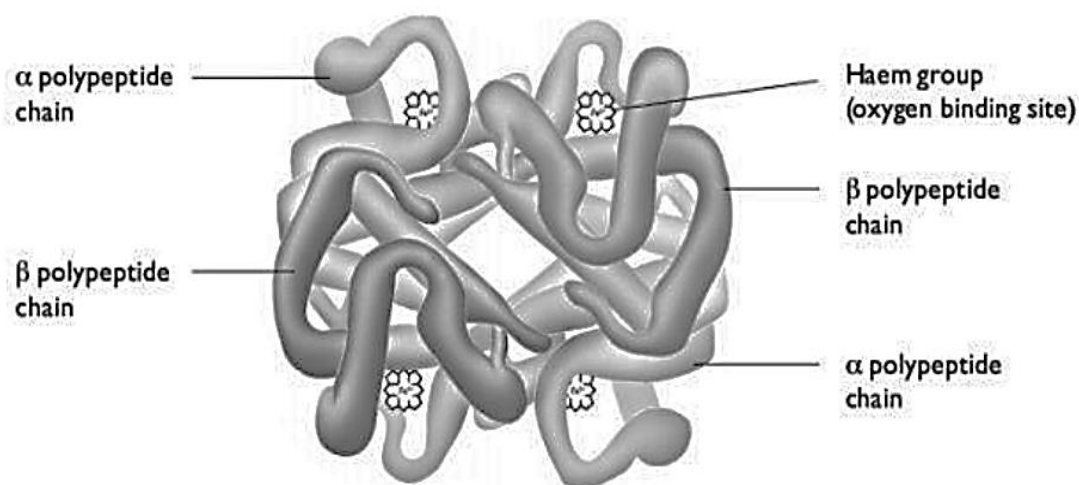
- ✚ In actively respiring tissues with high CO_2 concentrations, Carbonic acid is formed which dissociates to form Hydrogen ions and Hydrogen carbonate ions.
- ✚ The hydrogen ions combine with haemoglobin to form haemoglobinic acid, HHb due to the nature of the polypeptide bonds.
- ✚ This causes the haemoglobin to release oxygen.
- ✚ By removing the hydrogen ions from solution, haemoglobin helps to maintain the pH of the blood close to neutral thus it acts as a buffer.

4. Ability to undergo Redox reactions

- ✚ The haem group contained within each polypeptide chain enables the haemoglobin molecule to combine with oxygen.
- ✚ Oxygen molecules can reduce Fe^{2+} , in the haem group.
- ✚ One oxygen molecule (two oxygen atoms) can combine with each haem group, so one haemoglobin molecule can combine with four oxygen molecules (eight oxygen atoms).

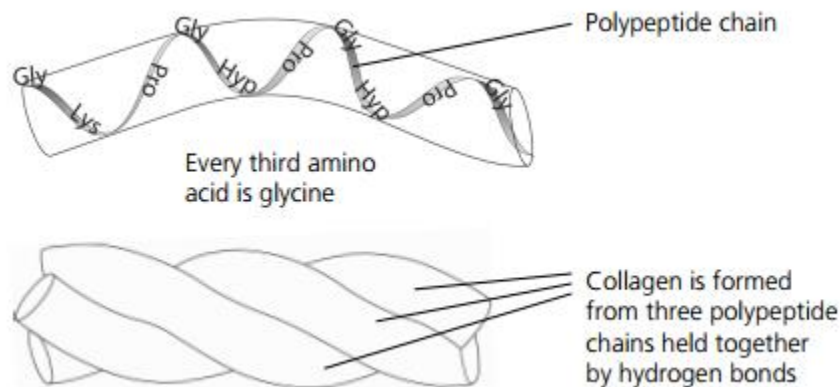
5. Ability to transport Oxygen

- ✚ The overall shape of the haemoglobin molecule enables it to pick up oxygen when the oxygen concentration is high, and to release oxygen when the oxygen concentration is low.
- ✚ Small changes in oxygen concentration have a large effect on how much oxygen the haemoglobin molecule can hold.
- ✚ Once one oxygen molecule has combined with one haem group, the whole molecule changes its shape in such a way that it is easier for oxygen to combine with the other three haem groups.



COLLAGEN

- ✚ It is an insoluble fibrous protein found in skin, tendons, cartilage and the walls of blood vessels.
- ✚ A collagen molecule consists of three polypeptide chains, each in the shape of a helix.
- ✚ These three helical polypeptides are wound around each other, forming a three-stranded 'rope' or 'triple helix'.
- ✚ The three strands are held together by hydrogen bonds and some covalent bonds.
- ✚ Almost every third amino acid in each polypeptide is glycine, the smallest amino acid.
- ✚ Glycine is found on the insides of the strands and its small size allows the three strands to lie close together to form a tight coil. And any other amino acid would be too large.
- ✚ Each complete, three-stranded molecule of collagen interacts with other collagen molecules running parallel to it and covalent bonds form between the R groups of amino acids lying next to each other.
- ✚ These cross-links hold many collagen molecules side by side, forming fibrils.
- ✚ The ends of the parallel molecules are spread out; if they were not, there would be a weak spot running right across the collagen fibril.
- ✚ Finally, many fibrils lie alongside each other, forming strong bundles called fibres.



RELATIONSHIP BETWEEN STRUCTURE AND FUNCTION IN COLLAGEN

1. Insolubility

- ✚ Collagen molecules are very long and are too large to be able to dissolve in water.

2. High tensile strength

- ✚ Three polypeptide chains wind around one another, held together by hydrogen bonds, to form a three-stranded molecule that can withstand quite high pulling forces without breaking.
- ✚ This structure also allows the molecules to stretch slightly when pulled.

3. Compactness

- ✚ Every third amino acid in each polypeptide is glycine, whose R group is just a single hydrogen molecule.
- ✚ Their small size allows the three polypeptide chains in a molecule to pack very tightly together.

4. Formation of fibres

- ✚ There are many lysine molecules in each polypeptide, facing outwards from the three-stranded molecule.
- ✚ This allows covalent bonds to form between the lysine R groups of different collagen molecules, causing them to associate to form fibres.

FUNCTIONS OF PROTEINS

1. Structural materials, body support and movement

- Proteins make up more than 50% dry mass in cells, they are an important part of membranes that surround the cells and cell organelles.
- Proteins form body tissues such as muscles, Nerve and Connective tissues.
- Body structures which are made up of the protein keratin include nails, wool and hair, feathers, Horns and hooves.
- Proteins such as myosin are responsible for the movement of muscles.

2. Transport

- Proteins are involved in the transport of substances in living organisms.
- For example, the protein Haemoglobin (iron containing protein found in red blood cells) transports oxygen from the lungs to other parts of the body.
- Other proteins transport molecules across cell membranes during active transport.

3. *Regulation of Body processes*

- Some proteins called messenger proteins, transmit signals between different cells, tissues and organs to co-ordinate biological processes. Many messenger proteins are hormones.
- The Hormone insulin regulates the concentration of sugar in the blood.
- The hormone secretin stimulates the pancreas and intestines to create digestive juices that help with the process of digestion.
- Proteins also form enzymes. Enzymes are specialised proteins that regulate the rate of chemical reactions in living organisms.

4. *Protection*

- Proteins help to protect the body against diseases and helps wounds to heal.
- Proteins form antibodies that attack invading bacteria and viruses to provide immunity from diseases. They identify and help destroy bacteria and viruses that can harm the body.
- Proteins are also needed for blood clotting. These proteins called fibrin, form a network of fibres that hold platelets in place so that blood can clot and form scab over a wound.

5. *Buffers*

- Buffer systems help to control the pH of a solution, for example blood.
- In the body proteins form buffer systems that help to regulate the acidity and alkalinity in and around cells.
- The protein Haemoglobin for example can bind to small amounts of acid in the blood and help to remove the acid.

KEY CONCEPT.

..... *WATER.*

OBJECTIVES.

..... *Describe the structure and properties of water.*
 *Explain the roles of water in living organisms and as an environment*

WATER

- ✚ Water is a major component of cells, typically forming between 70% and 95% of the mass of the cell.
- ✚ It provides a habitat for those organisms that live in water.
- ✚ Because water is a liquid, it provides a medium for molecules and ions to mix in, and hence a medium in which life can evolve.
- ✚ As a small molecule, it would exist as a gas at normal Earth temperatures were it not for its special property of cohesion by Hydrogen bonding making the molecules more difficult to separate.

UNIQUE PROPERTIES OF WATER

1. *Water is a solvent*

- ✚ Water is an excellent solvent for ions and polar molecules because the water molecules are attracted to ions and polar molecules and therefore collect around them and separate them.
- ✚ Once a chemical is in solution, it is free to move about and react with other chemicals.
- ✚ Most metabolic processes in living organisms take place in solution.
- ✚ Non-polar molecules such as lipids are insoluble in water and, if surrounded by water, tend to be pushed together by the water, since the water molecules are attracted to each other.

2. *High specific heat capacity*

- ✚ The heat capacity of a substance is the amount of heat required to raise its temperature by a given degree.
- ✚ Water has a relatively high heat capacity.
- ✚ In order for the temperature of a liquid to be raised, the molecules must gain energy and move about more rapidly.
- ✚ The hydrogen bonds make it difficult for molecules to move about freely and this explains why more energy is needed to raise the temperature of water.
- ✚ The high heat capacity of water has important biological implications because it makes water more resistant to changes in temperature.
- ✚ This means that the temperature within cells and within the bodies of organisms (which have a high proportion of water) tends to be more constant than that of the air around them.
- ✚ Biochemical reactions therefore operate at relatively constant rates and are less likely to be adversely affected by extremes of temperature.
- ✚ It also means that large bodies of water such as lakes and oceans are slow to change temperature as environmental temperature changes.
- ✚ As a result, they provide more stable habitats for aquatic organisms.

3. *High Latent heat of vaporization*

- ✚ The latent heat of vaporization is a measure of the heat energy needed to vaporise a liquid and cause it to evaporate.
- ✚ Water has a relatively high latent heat of vaporization.
- ✚ This is a consequence of its high heat capacity.
- ✚ The fact that water molecules undergo cohesion by hydrogen bonds means that large amounts of energy are needed for vaporization to occur, because hydrogen bonds have to be broken before molecules can escape as a gas.
- ✚ The energy transferred to water molecules during vaporization results in a corresponding loss of energy from their surroundings, which therefore cool down.
- ✚ This is biologically important because it means that living organisms can use evaporation as a cooling mechanism, as in sweating or panting in mammals.
- ✚ A large amount of heat energy can be lost for relatively little loss of water, reducing the risk of dehydration.

4. *Lower Density in the solid state*

- ✚ Water is an unusual chemical because the solid form, ice, is less dense than its liquid form.
- ✚ Below 4 °C, the density of water starts to decrease.
- ✚ Ice therefore floats on liquid water and insulates the water under it.

- ✚ This reduces the tendency for large bodies of water to freeze completely, and increases the chances of life surviving in cold conditions.
- ✚ Changes in the density of water with temperature cause currents, which help to maintain the circulation of nutrients in the oceans.

5. Reagent

- ✚ Water takes part as a reagent in some chemical reactions inside cells.
- ✚ For example, it is used as a reagent in photosynthesis.
- ✚ Water is also essential for all hydrolysis reactions.
- ✚ Hydrolysis is the mechanism by which large molecules are broken down to smaller molecules, as in digestion.

6. High surface tension and Cohesion

- ✚ Water molecules have very high cohesion as they tend to stick to each other.
- ✚ This explains why water can move in long, unbroken columns through the vascular tissue in plants, and is an important property in cells.
- ✚ High cohesion also results in high surface tension at the surface of water.
- ✚ This allows certain small organisms, such as pond skaters, to exploit the surface of water as a habitat, allowing them to settle on or skate over its surface.

Qn. Explain the role of water in living organisms as a habitat

[8]

Water is an excellent solvent for ions and polar molecules because the water molecules are attracted to ions and polar molecules and therefore collect around them and separate them. This is an important feature of water as a habitat as it allows nutrients to dissolve and be taken up by aquatic organisms. It also allows gases like Oxygen to dissolve in water allowing aquatic organisms to respire.

Water has a relatively high heat capacity because the hydrogen bonds make it difficult for water molecules to move about freely. The high heat capacity of water has important biological implications because it makes water more resistant to changes in temperature. This means that large bodies of water such as lakes and oceans are slow to change temperature as environmental temperature changes. As a result, they provide more stable habitats for aquatic organisms. Biochemical reactions therefore operate at relatively constant rates and are less likely to be adversely affected by extremes of temperature.

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Water molecules have very high cohesion as they tend to stick to each other. High cohesion results in high surface tension at the surface of water and this allows certain small organisms, such as pond skaters, to exploit the surface of water as a habitat, allowing them to settle on or skate over its surface. It also allows the Jesus lizard to run on the water surface.

“Time consumes everything including History and Literature. But the realm of science shall forever exist”.

Joy D Moyo-2019.